

Attack–Fault Identifiability and the Cost of Ambiguity at the Electronic Trading Gateway: An Exact Criterion, Its Fragility under State Feedback, and a Retrospective Simulation Study

Chiheb Mejrissi^{a,*}

^aDepartment of Computer Science, Example University, 1 Example Street, Example City, Country

Abstract

Trading-gateway operators must respond to suspicious order flow whose cause, a cyber compromise or a software fault, they may be unable to establish. We ask what gateway observations can reveal about the cause, and what the ambiguity costs. For a linear-Gaussian model of gateway-observable order flow, we prove that a compromise and a fault are observationally indistinguishable if and only if their projected injections coincide, and that under this confounding no observation-based policy beats the best constant action, so ambiguity has a strictly positive cost under stated cost conditions, also under approximate confounding. When the incident propagates through the dynamics, confounding survives only if the injection difference is Kalman-unobservable, which fails generically. We then study retrospectively, with complete-episode features, ABIDES simulator designs that match compromise and fault on chosen summary statistics; they do not meet the theorem’s premise, so this is not its test. Calibration by search did not bring the summary-matched regime to chance in any design (clearly in three, marginally in Study 5, accuracy lower bound 0.525); in some it was more separable than contrast. With a fault loss modelled from measured submitted volume, a classifier-based policy is never worse than the best constant policy with a supported interval, under both fault-cost conventions; its improvement is supported at every prior in 6 of 10 profile–regime units and at

*Corresponding author.

Email address: `chiheb.mejrissi@example.org` (Chiheb Mejrissi)

none in Study 5’s summary-matched unit (per-cell intervals, unadjusted for multiplicity). Costs are illustrative market-impact reconstructions, and no result measures the value of acting at a decision deadline.

Keywords: cyber-physical security, electronic trading, attack–fault identifiability, cost of ambiguity, decision under uncertainty, incident response, agent-based market simulation

1. Introduction

A trading participant can generate harmful or suspicious orders because of a compromised credential, a software defect, its own order-generation policy, ordinary market conditions, or a combination of these. An operator — the incident-response controller of a broker or trading firm — must often act before the cause is known: throttle, cancel orders, revoke access, or isolate a process. Each action has a cost when applied to a case that did not need it, and each carries risk when withheld from a case that did. Before asking when to act, one must know what the evidence can reveal at all and what the residual ambiguity costs. This paper answers those two prior questions. Its results concern the laws of whole observation trajectories and, empirically, retrospective diagnosis from complete episodes with a hypothetical decision cost attached; how much of that value is available at an operational decision deadline would need prefix features and a pre-committed cutoff, and is not answered here (Section 3.5).

Two questions precede any policy question. First: **can the causes be told apart** from the evidence an operator is ordinarily permitted to observe (order acknowledgments, market data, session and process telemetry)? Second, if not: **how costly is that ambiguity**, measured against the best action an operator could choose without the observation? A threshold-based kill switch answers the second question without the first: it trips on an order rate, a cancel ratio or a loss limit and acts, with no attempt to establish cause.

This paper answers the first question with a formal identifiability result (Sections 3–4), scoped to a linear-Gaussian order-flow model, as is the closest structural analog, the literature distinguishing cyberattacks from physical faults in cyber-physical systems. Section 4.1 shows that this confounding is fragile within the linear model itself once the incident propagates through the dynamics, through an exact connection to classical observability. Sec-

tions 5–6 check the theory in a linear pilot and define the cost of ambiguity. Section 7 studies retrospectively, in the ABIDES agent-based market simulator, what matching two incident mechanisms on chosen summary statistics does to classification and to decision cost, with features computed on complete episodes. It is not a test of Theorem 1: the constructions were never shown to satisfy its premise, equality of the full observation-trajectory law.

Contributions.

1. In the linear-Gaussian order-flow model, a compromise and a fault are observationally indistinguishable iff their projected injections coincide (Theorem 1); fault-mimicking needs the projected compromise direction to lie in the span of the projected fault direction, and rank-two separation gives distinguishability, not high accuracy (Corollaries 1–2, with a Gaussian discriminability benchmark and its unequal-prior ambiguity cost, Proposition 3). Under confounding the cost of ambiguity is strictly positive whenever harm exceeds routine containment cost, disruption is costly and the prior lies strictly between 0 and 1, with a total-variation extension to approximate confounding (Propositions 1–2). A linear pilot is consistent with these predictions (Sections 5–6).
2. Confounding that survives one-shot observation does not survive state feedback unless it lies in the Kalman unobservable subspace; generically it is fragile (Theorem 2, its finite-horizon version Theorem 3, and Corollary 3). In a simulation, an exactly confounded pair becomes classifiable under generic coupled dynamics while it stays at chance under output-level injection.
3. An ABIDES study of observation matching (Section 7). In the four tested injection designs (one simulator, one shared infrastructure), calibration by search did not bring the summary-matched regime to chance, marginally so in the Study 5 profile, and in some designs the summary-matched regime was more separable than the contrast regime. On the paired Era-3 basis (episodes generated with the hardened simulation pipeline, Section 7.4), the classifier-based policy is never worse than the best constant policy with a supported interval, under both fault-cost conventions; where it improves on it, and how its comparison with always-contain depends on the fault-cost convention, is stated in Section 7.3.

Section 7.5 states, as an outlook, the question these cells leave open.

2. Related work

Decision-theoretic cyber response. Belief-state (POMDP) formulations of cyber defense track an attacker’s progression through a network and choose actions under noisy alerts [1]; related work formulates response as a Stackelberg game over attack-response trees [2], or balances service continuity against intrusion risk under partial observability [3], with learning-based extensions against dynamic attackers and under model misspecification [4, 5], and model-based decision support [6]. None of this literature prices market-impact costs of the kind Section 6 constructs; our own cost structure is a loss table with a market-impact proxy, not a model of endogenous liquidity replacement.

Cyber-physical fault-versus-attack differentiation. The closest structural analog to our question is the literature distinguishing physical faults from cyberattacks in power and other cyber-physical systems. Pasqualetti, Dörfler and Bullo [7] characterize, with geometric control theory, which attacks are undetectable or unidentifiable by an output-based monitor (those that excite only the zero dynamics); Theorem 2 uses the attack-free special case of this object, the Kalman unobservable subspace. Roy and Dey [8] estimate an anomalous input to a linear system with a sliding-mode unknown-input observer and derive a closed-form distinguishability criterion between fault and attack, including a fault-mimicking condition that is the direct analog of our Corollary 1. Observer-based schemes that detect and isolate faults and deception attacks simultaneously [9], and detectors designed to keep faults and attacks distinguishable in interconnected systems [10], address the design side of the same question; neither prices the response or treats trading. On the data-driven side, Anwar, Mahmood and Shah [11] separate simulated faults from attacks on an IEEE bus test system with a PCA projection and a support-vector classifier, and later systems fuse network and process data for attack-versus-failure attribution [12]. Closest to our decision-theoretic framing, a triage framework for energy cyber-physical systems attributes anomalies to physical faults or to measurement-integrity loss, prices misattribution with response-cost matrices against blanket responses, and observes that exactly matched fault/attack pairs are discriminated at chance [13]. That work is empirical and set in process and wind-farm control; it derives no necessary-and-sufficient indistinguishability condition, no cost floor under confounding and no state-feedback fragility, and does not address trading. Its matched-pair observation is the empirical counterpart of

our Theorem 1 and Proposition 1, whereas Section 7 concerns pairs matched only on summary statistics.

Roy and Dey validate their criterion in simulations of the same linear model it is proved for, as does our pilot (Section 5). Section 7 attempts a nonlinear construction in a multi-agent market and does not claim to test their criterion or ours. Stealth that holds in a linear model need not hold in a nonlinear one: stealthy injections must be redesigned for nonlinear differential-algebraic grid models [14]. Alisic and Amin [15] extend sensor-attack analysis to dynamics on Lie groups and show that an attack keeps its impact and stealth across operating conditions only if it commutes with the flow. No analogous condition has been derived for the order-flow model of Section 3, so we claim neither that our market satisfies it nor that it fails it.

Market manipulation detection and trading kill switches. A large literature detects manipulative patterns (spoofing, wash trading) from trade and order-book data, for example with supervised and unsupervised deep-learning detectors [16]; it classifies trading patterns as manipulative or not and does not ask whether a compromised gateway and a faulty one can be told apart, nor price the response. Manipulative order generation can be automated through compromised brokerage accounts [17], which supports modelling a compromise as a structured perturbation of order flow. A threshold trigger on an order-flow statistic is an observation-dependent policy; the constant policies of this paper (always contain, always leave) are decision-theoretic reference points, the best achievable without any observation, not models of kill switches.

Costly information acquisition. Models of costly information acquisition in asset markets [18] and optimal-stopping formulations of costly sequential acquisition [19] concern traders’ incentives to learn about asset value, not an operator’s decision to verify an incident’s cause; the question matters to the companion paper rather than to this one.

Scope of the contribution. The cyber-physical distinguishability literature [8, 7, 11], cost-aware triage [13] and the Lie-group transfer analysis [15] share substantial machinery with Sections 3–4. What we add is (1) an exact identifiability criterion and cost floor for the compromise-versus-fault question at a trading gateway, with a market-impact loss table rather than a detection-accuracy metric; (2) a comparison, through the unobservable subspace, of output-level and state-level injection of the same incident pair, showing that exact output-level confounding is generically not preserved once the incident propagates (Theorem 2); and (3) an empirical study of whether

summary-matching produces confounding in a nonlinear multi-agent simulator. Items (1) and (2) are new results for this question obtained with established tools, not new mathematical machinery; (3) is a retrospective study in one simulator, not a test of (1). The prior-shift threshold correction of Section 6 is standard [20, 21].

3. Model

Throughout, K (“compromise”) and F (“fault”) name fully specified perturbations of the order-flow process — a directional bias, an elevated order rate or an altered cancellation pattern, with a chosen severity and onset — not claims about real attackers or real defects. The results concern *behavioural-mechanism identification*, whether two specified generative processes can be told apart from gateway-observable order flow, not inference about intent or authorization, which would need other evidence (credential logs, code-change records). The four ABIDES designs of Section 7 are parameterizations of one shared simulation infrastructure, not independent implementations.

3.1. Order-flow dynamics

Let $x_t \in \mathbb{R}^n$ be a latent representation of normal order-flow state (for example, components loading onto price pressure, volume intensity and cancellation propensity), evolving under known linear-Gaussian dynamics with transition matrix A and process-noise covariance Q . The complete model is Definition D2 below.

3.2. Incident injection

At an onset time τ , possibly unknown to the operator, an incident of mechanism $M \in \{K, F\}$ injects an additive perturbation into the pre-observation order-flow signal, with direction b_M and severity θ_M .

Definition D1 (Mechanism instance). A mechanism instance is a triple (M, θ_M, b_M) with cause $M \in \{K, F\}$ (K : compromise, F : fault), a deterministic severity $\theta_M \in \mathbb{R}$ and a deterministic injection direction $b_M \in \mathbb{R}^n$. Statements compare one fixed instance of K with one fixed instance of F (pairwise-instance reading); no distribution over severities, directions or onsets within a cause is modelled.

3.3. Gateway-observable output

The operator observes a lower-dimensional ($m < n$), noisy projection of order flow through a fixed, known matrix $C \in \mathbb{R}^{m \times n}$, representing what a gateway or market-data feed surfaces (spread, depth, cancellation rate). In this model the incident shifts the signal at the point of observation only; Section 4.1 lets it propagate through the dynamics.

Definition D2 (Output-level injection model). Fix n, m , a matrix $A \in \mathbb{R}^{n \times n}$, an observation matrix $C \in \mathbb{R}^{m \times n}$, covariance matrices $Q \in \mathbb{R}^{n \times n}$ and $\Sigma \in \mathbb{R}^{m \times m}$, and an onset time τ . Under a mechanism instance (M, θ_M, b_M) (Definition D1), the latent state, the perturbed state and the observation evolve for $t \geq 0$ as

$$\begin{aligned} x_{t+1} &= Ax_t + w_t, & \tilde{x}_t &= x_t + \mathbb{I}[t \geq \tau] \theta_M b_M, \\ y_t &= C\tilde{x}_t + \varepsilon_t, \end{aligned}$$

with $w_t \sim \mathcal{N}(0, Q)$ and $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$. The injection enters the observation only; the state recursion does not see it. (Section 3 additionally takes $m < n$; no statement below uses that restriction.)

Everything except the injection term is common to the two causes. Section 4 rests on the following standing assumptions, which the implementation satisfies; the finite-horizon experiments of Sections 5–6 always include time steps after the onset, so (A5) holds.

Standing assumptions.

(A1) Gaussian, independent noise. $w_t \sim \mathcal{N}(0, Q)$ and $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$, with Q and Σ symmetric positive semidefinite; the family $\{w_t\}_{t \geq 0} \cup \{\varepsilon_t\}_{t \geq 0}$ is mutually independent (across time and across the two sequences), and its law does not depend on the instance.

(A2) Deterministic initial state. $x_0 = 0$. (The proofs use only that x_0 is Gaussian, possibly degenerate, independent of the noise, with a law that does not depend on the instance; $x_0 = 0$ is the implemented special case.)

(A3) Known, common model. A, Q, C, Σ are known and identical under both causes; only the injection term $\theta_M b_M$ depends on the instance.

(A4) Common onset. The onset τ is deterministic, common to both causes, not necessarily known to the operator. (The proofs use only that τ is deterministic and common; no statement requires a policy to know τ .)

(A5) Window reaches the onset. The observation window W contains at least one $t \geq \tau$. Theorem 2 uses the stronger form $W \supseteq \{t : t \geq \tau\}$ and Theorem 3 the stronger form $W \supseteq \{\tau, \dots, \tau + H - 1\}$.

(A6) Positive-definite observation noise. $\Sigma \succ 0$. Used only to apply Proposition 3 to the model on a finite window: it makes the covariance of the stacked observations positive definite ($V \succ 0$). No other statement uses a positivity condition on Σ or Q ; Q is only required to be positive semidefinite.

Definition D3 (Observation window; observational indistinguishability). An observation window is a set $W \subseteq \{0, 1, 2, \dots\}$ of time indices, finite or not; the observation trajectory on W is $Y_W = (y_t)_{t \in W}$. Write P_M^W for the law of Y_W under instance (M, θ_M, b_M) (for infinite W , the law on the product σ -algebra, determined by its finite-dimensional marginals). Instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally indistinguishable on W if $P_K^W = P_F^W$. Theorem 1 uses $W = \{t : t \geq 0\}$; “indistinguishable at every $t \geq \tau$ ” (Theorem 2) means $P_K^W = P_F^W$ for $W = \{t : t \geq \tau\}$; Theorem 3 (the finite-horizon version of Theorem 2) uses $W = \{\tau, \dots, \tau + H - 1\}$.

Definition D4 (Distinguishable in the population limit). Two instances are distinguishable in the population limit on W if they are not observationally indistinguishable on W , i.e. $P_K^W \neq P_F^W$ (equivalently $\text{TV}(P_K^W, P_F^W) > 0$, Definition D8). This is a statement about the two laws only; it asserts nothing about the accuracy attainable from finitely many observations.

Definition D5 (Gaussian discriminability). If $Y \mid M \sim \mathcal{N}(\mu_M, V)$, $M \in \{K, F\}$, with a common covariance $V \succ 0$, let $\delta = \mu_K - \mu_F$ and $d = \sqrt{\delta^\top V^{-1} \delta} \geq 0$ (the whitened mean separation).

3.4. The loss table, and what it assumes

The cost analysis uses the two-action, two-cause loss table

Action	Compromise K	Fault F
Contain	C_{OP}	DISRUPT
Leave alone	HARM	0

a special case of the general table below; Proposition 1 keeps these symbols.

Definition D6 (Loss table). The general two-action, two-cause loss table $L(a, M)$, $a \in \{\text{contain, leave alone}\}$, $M \in \{K, F\}$, is

$$\begin{aligned} L(\text{contain}, K) &= C_K, & L(\text{contain}, F) &= C_F, \\ L(\text{leave alone}, K) &= H_K, & L(\text{leave alone}, F) &= H_F, \end{aligned}$$

with $C_K, C_F, H_K, H_F \in \mathbb{R}$. This paper’s table (Section 3) is the special case

$$C_{OP} = C_K, \quad \text{DISRUPT} = C_F, \quad \text{HARM} = H_K, \quad H_F = 0.$$

Propositions 1 and 2 are stated in this paper’s symbols C_{OP} , DISRUPT, HARM.

Definition D7 (Prior, policies, Bayes risk, oracle risk, cost of ambiguity). The cause is drawn as $M = K$ with prior probability $p = P(K)$ and $M = F$ with probability $1 - p$; given M , the observation trajectory Y_W has law P_M^W . A policy is a measurable map π from observation trajectories on W to $[0, 1]$, read as the probability of the action *contain* (randomized policies allowed; a deterministic policy takes values in $\{0, 1\}$). Its expected cost is

$$\begin{aligned} R(\pi) = & p \mathbb{E}_K [\pi(Y_W) C_K + (1 - \pi(Y_W)) H_K] \\ & + (1 - p) \mathbb{E}_F [\pi(Y_W) C_F + (1 - \pi(Y_W)) H_F]. \end{aligned}$$

The best achievable expected cost is $R^* = \inf_{\pi} R(\pi)$; the oracle, which observes M , has expected cost $R_{\text{oracle}} = p \min\{C_K, H_K\} + (1 - p) \min\{C_F, H_F\}$; the cost of ambiguity is $\text{CoA}^* = R^* - R_{\text{oracle}}$. Where the observation is the raw trajectory Y the subscript Y is written (R_Y^* , CoA_Y^*).

Definition D8 (Observation densities; total variation). Let μ be a σ -finite measure dominating P_K and P_F (e.g. the finite measure $P_K + P_F$), and f_K, f_F their densities with respect to μ ; $\int \cdot dy$ denotes integration against μ . The total variation distance is

$$\text{TV}(P, P') = \sup_B |P(B) - P'(B)| = \frac{1}{2} \int |f - f'| dy,$$

so that $1 - \text{TV}(P_K, P_F) = \int \min\{f_K, f_F\} dy$.

The table encodes a scenario in which leaving a fault alone is free, for example a logging error that inflates apparent order counts. The strict value of distinguishing the two causes is zero whenever one action is optimal for both; Proposition 1’s cost floor is a property of loss structures in which the causes call for different actions. The simulated fault mechanisms of Section 7 submit real orders, so Section 6.2 replaces the zero entry by a positive fault loss H_F , modelled from the submitted volume measured on the simulated episodes. The per-design audit of the fault mechanisms is in Supplement S4.

3.5. Observation timing and scope

Section 1 places the paper’s questions before the timing question of when to act; this section states what that leaves out. Theorem 1 and Proposition 1

concern the whole observation trajectory, and both the linear pilot and the ABIDES study use whole-episode summary features (mean, standard deviation and net drift of each observed channel), not a prefix available at a pre-committed decision deadline. The study is therefore a retrospective diagnosis study: nothing enforces that features use only information available when an operator would have to act. A real-time claim would need an explicit observation cutoff and prefix features; we do not make one. This does not affect the identifiability results, which concern the laws of whole trajectories, but the costs of Sections 6–7 do not measure the value of information at an operational decision point.

4. Theoretical results

Proofs of Theorem 1, Proposition 1 and Theorem 2 are in Appendix Appendix A; the proofs of all statements, one per statement, are in Supplement S1, submitted with the manuscript.

Theorem 1 (Indistinguishability criterion). Under the model of Section 3, with common onset time τ , mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) induce identical distributions over the observation trajectory $\{y_t\}_{t \geq 0}$ if and only if

$$\theta_K C b_K = \theta_F C b_F. \quad (*)$$

Theorem 1 uses Definition D2, Definition D1, Definition D3 (window $W = \{t : t \geq 0\}$) and assumptions (A1)–(A5).

No rank, observability or definiteness condition on A , C , Q or Σ is needed, and the restriction $m < n$ of Section 3 is not used; the onset τ may be any deterministic time.

Corollary 1 (Fault-mimicking compromise). Fix $u = C b_K$, $v = C b_F$, and a nonzero compromise severity $\theta_K \neq 0$. If the fault severity θ_F may be any real number, then

$$\exists \theta_F \in \mathbb{R} : \theta_K u = \theta_F v \iff u \in \text{span}(v).$$

Corollary 1 uses Definition D1 and condition $(*)$ of Theorem 1; the equivalence itself is linear algebra and uses no probabilistic assumption.

Two precisions. Linear dependence of u and v means rank at most one, and its subcases differ: if $v = 0$ but $u \neq 0$ no θ_F solves the equation, and if both vanish every θ_F does; with positive severities, nonzero u, v must lie on

the same ray. All results compare one fixed instance of each mechanism (Definition D1): families or superpositions of fault directions, and laws averaged over severities or onsets, would need a model this paper does not specify.

Corollary 2 (Separation implies distinguishability, not high accuracy). If $\text{rank}([Cb_K \ Cb_F]) = 2$, no pair (θ_K, θ_F) other than $(0, 0)$ satisfies $(*)$; for any instance pair with $(\theta_K, \theta_F) \neq (0, 0)$, K and F are then distinguishable in the population limit.

Corollary 2 uses Definition D1, Definition D3, Definition D4 and, through Theorem 1, Definition D2 and assumptions (A1)–(A5).

Separation does not imply good finite-sample classification: the projected directions can be small relative to the noise. The benchmark is the whitened separation d of Definition D5; under equal priors the Bayes-optimal accuracy and area under the ROC curve (AUC) are the standard identities [22, 23]

$$\text{Acc}^* = \Phi(d/2), \quad \text{AUC}^* = \Phi(d/\sqrt{2}).$$

A warning on notation. Writing $Z = \phi(Y)$ for the engineered features supplied to a classifier, $R_{\text{oracle}} \leq R_Y^* \leq R_Z^* \leq R_{\text{const}}$, and a feature ablation moves R_Z^* , not R_Y^* . The empirical sections concern R_Z^* ; the two coincide only when ϕ is sufficient, which is not established for any feature set used here.

Proposition 1 (Cost of ambiguity is bounded below under confounding). Suppose $(*)$ holds (the confounded case), $0 < p < 1$, DISRUPT > 0 , and HARM $> C_{OP}$ (harm strictly worse than routine containment). Then the best achievable expected cost over the policies of Definition D7 (randomized policies allowed) is

$$R^* = \min\{p C_{OP} + (1 - p) \text{DISRUPT}, \ p \text{HARM}\},$$

the oracle’s expected cost is $R_{\text{oracle}} = p C_{OP}$, and

$$\boxed{\text{CoA}^* = R^* - R_{\text{oracle}} = \min\{(1 - p) \text{DISRUPT}, \ p(\text{HARM} - C_{OP})\} > 0.}$$

Proposition 1 uses Definition D6 (this paper’s symbols), Definition D7 and, through Theorem 1 (to pass from $(*)$ to $P_K^W = P_F^W$), Definition D2, Definition D1, Definition D3 and assumptions (A1)–(A5).

Remark. The hypothesis HARM $> C_{OP}$ is essential for the strict inequality: if HARM $\leq C_{OP}$, leaving alone is optimal under both causes,

the best constant action attains the oracle’s cost and $\text{CoA}^\star = 0$. Likewise $\text{CoA}^\star = 0$ when $p \in \{0, 1\}$ or $\text{DISRUPT} = 0$.

For a general finite loss table $L(a, M)$ the cost of ambiguity is

$$\min_a \{p L(a, K) + (1 - p)L(a, F)\} - p \min_a L(a, K) - (1 - p) \min_a L(a, F),$$

strictly positive exactly when no single action is optimal for both causes at positive probability. Exact equality is a brittle operational target, so we extend the bound to approximate confounding.

Proposition 2 (Total-variation extension of Proposition 1). Let the loss table and prior be as in Proposition 1, with $0 < p < 1$, $\text{DISRUPT} > 0$ and $\text{HARM} > C_{OP}$, but do not assume (*). Let f_K, f_F be the observation densities (Definition D8) and set $a = p(\text{HARM} - C_{OP})$, $b = (1 - p) \text{DISRUPT}$. Then

$$\text{CoA}^\star = \int \min\{af_K(y), bf_F(y)\} dy = \frac{a + b - \int |af_K - bf_F| dy}{2},$$

$$\min(a, b) (1 - \text{TV}(P_K, P_F)) \leq \text{CoA}^\star \leq \min(a, b),$$

and, if a common surrogate law $\bar{P}$ satisfies $\text{TV}(P_K, \bar{P}) \leq \epsilon_K$ and $\text{TV}(P_F, \bar{P}) \leq \epsilon_F$, then

$$\text{CoA}^\star \geq \min(a, b) \max\{0, 1 - \epsilon_K - \epsilon_F\}.$$

Proposition 2 uses Definition D6 (this paper’s symbols), Definition D7 and Definition D8; it uses none of (A1)–(A6): it holds for any two observation laws.

The first identity is the classical relation between Bayes risk and total variation [24]; the lower bound recovers Proposition 1 when $\text{TV} = 0$. The surrogate-law bound states the premise any “approximate confounding transfers approximately” argument needs: a bound on distributional approximation error, not a match on one statistic. A classifier performing near chance cannot certify a small total variation distance.

For the equal-covariance Gaussian setting, likelihood-ratio integration gives the population-optimal ambiguity cost as a function of d , the unequal-prior cost analog of Corollary 2; it is a standard consequence of Gaussian testing, and a summary-based classifier can only do worse.

Proposition 3 (Gaussian unequal-prior ambiguity cost). Let $Y | M \sim \mathcal{N}(\mu_M, V)$, $M \in \{K, F\}$, with common covariance $V \succ 0$ and discriminability $d > 0$ (Definition D5), prior $p = P(K)$, and the general loss

table with $a = p(H_K - C_K)$, $b = (1 - p)(C_F - H_F)$, $a, b > 0$. Then the population-optimal ambiguity cost is

$$\text{CoA}_Y^*(d) = a \Phi\left(\frac{\log(b/a)}{d} - \frac{d}{2}\right) + b \Phi\left(\frac{\log(a/b)}{d} - \frac{d}{2}\right),$$

which tends to $\min(a, b)$ as $d \rightarrow 0^+$ (the value at $d = 0$, exact confounding, recovering Proposition 1, is obtained as this limit) and decreases toward 0 as $d \rightarrow \infty$.

Proposition 3 uses Definition D6 (general symbols), Definition D7, Definition D8 and Definition D5; applied to the model on a finite window it needs (A1)–(A5) (common covariance) and (A6) ($V \succ 0$).

4.1. Confounding is fragile within the linear model: state feedback

Theorem 1 models the incident as an output-level disturbance: it shifts x_t only at the point of observation, while the recursion $x_{t+1} = Ax_t + w_t$ evolves as if nothing happened. A compromise or a persistent fault behaves more like a state-level disturbance: behaviour at t feeds into the state at $t + 1$, as an incident agent's orders feed into the book that other agents react to. This changes the confounding condition.

Definition D9 (State-level injection model). As Definition D2, except that the perturbed state drives the recursion from the onset on:

$$\begin{aligned} x_{t+1} &= Ax_t + w_t \quad (t < \tau), & x_{t+1} &= A\tilde{x}_t + w_t \quad (t \geq \tau), \\ \tilde{x}_t &= x_t + \mathbb{I}[t \geq \tau] \theta_M b_M, & y_t &= C\tilde{x}_t + \varepsilon_t. \end{aligned}$$

The injection is persistent (added at every $t \geq \tau$) and propagates through A .

Definition D10 (Unobservable subspace; observability). The (Kalman) unobservable subspace of (A, C) is $\mathcal{N}(A, C) = \bigcap_{k=0}^{n-1} \ker(CA^k)$. The pair (A, C) is observable if $\mathcal{N}(A, C) = \{0\}$, equivalently if the stacked matrix

$$\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

has rank n .

Theorem 2 (State-feedback confounding requires unobservability). Under the state-level injection model, let $\Delta_0 = \theta_K b_K - \theta_F b_F$. Mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally indistinguishable at every $t \geq \tau$ if and only if Δ_0 lies in the unobservable subspace of the pair (A, C) , i.e.

$$\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k),$$

the classical (Kalman) unobservable subspace of (A, C) , where n is the state dimension.

Theorem 2 uses Definition D9, Definition D1, Definition D3 (window $W = \{t : t \geq \tau\}$), Definition D10 and assumptions (A1)–(A4) with (A5) in its all-time form.

Theorem 3 (Finite-horizon version of Theorem 2). Under the state-level injection model, let $\Delta_0 = \theta_K b_K - \theta_F b_F$, let $H \geq 1$ be the number of post-onset observations, and define

$$G_H = \begin{bmatrix} C \\ C(I + A) \\ \vdots \\ C \sum_{j=0}^{H-1} A^j \end{bmatrix}.$$

Mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) induce the same law of the observations over the window of H post-onset observations if and only if $G_H \Delta_0 = 0$, which holds if and only if $CA^j \Delta_0 = 0$ for $j = 0, \dots, H-1$. If $H \geq n$, this is equivalent to the all-time condition of Theorem 2, $\Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k)$.

Theorem 3 uses Definition D9, Definition D1, Definition D3 (window $W = \{\tau, \dots, \tau + H - 1\}$), Definition D10 and assumptions (A1)–(A4) with (A5) in its finite-horizon form.

The result is not inherently asymptotic: for $H \geq n$, exact equality over H post-onset observations already forces the all-time Kalman condition. What depends on horizon and noise is finite-sample discriminability when the condition fails, which Theorem 2 does not bound.

Corollary 3 (State-feedback confounding under observability). Under the hypotheses of Theorem 2, if (A, C) is observable, then mechanism instances (K, θ_K, b_K) and (F, θ_F, b_F) are observationally indistinguishable at every $t \geq \tau$ if and only if $\theta_K b_K = \theta_F b_F$.

Corollary 3 uses everything Theorem 2 uses, and Definition D10; its accompanying remark also uses Definition D11.

Definition D11 (Generic). For fixed C , a property of A holds generically if it holds for Lebesgue-almost every $A \in \mathbb{R}^{n \times n}$, i.e. it fails only on a Lebesgue-null subset of $\mathbb{R}^{n \times n}$.

Remark (Theorem 1’s confounded pairs are generically state-feedback-fragile). Theorem 1 only requires $\Delta_0 \in \ker(C)$ — a single condition. Theorem 2 requires Δ_0 to survive n such conditions simultaneously ($\ker(C), \ker(CA), \dots, \ker(CA^{n-1})$). Whenever (A, C) is observable (for fixed $C \neq 0$, the generic case for a randomly chosen A), the unobservable subspace is $\{0\}$, so **the only state-feedback-robust confounding is the trivial one where $\theta_K b_K = \theta_F b_F$ exactly as vectors** — i.e., the two mechanisms must push the identical linear combination of latent state variables, not merely produce the identical *projection* under C .

The pilot’s construction (Section 5), which uses the null space of C to make $b_K \neq b_F$ with $Cb_K = Cb_F$, is exactly such a pair: $\Delta_0 \in \ker(C)$ but $\theta_K b_K \neq \theta_F b_F$. By Corollary 3 it cannot remain confounded under state feedback whenever (A, C) is observable, which for fixed $C \neq 0$ holds for Lebesgue-almost every A . With the pilot’s A , a scalar multiple of the identity, $\ker(C)$ is trivially A -invariant, a measure-zero case in which the pair stays confounded. We re-ran the pilot with a stable, coupled and observable A . Under state-level injection with generic coupled dynamics, classification accuracy rises with injection magnitude, while under output-level injection it stays at chance throughout; with scalar (uncoupled) dynamics the state-level case also stays at chance. Over 16 magnitudes and 10 seeds, coupled state-level accuracy rises from 0.628 to 1.000, while under output-level injection with the same coupled dynamics it stays between 0.497 and 0.503 (Supplement S2). These classifiers use whole-episode features; how quickly the difference becomes detectable after onset is not measured here (Supplement S2).

Two qualifications. “Generic” concerns a full-measure set of matrices; real, structured market dynamics can have unobservable components by construction, and we have not established that they behave like a generic matrix. A one-time initial-state perturbation and persistent forcing produce different mean-difference sequences but share the same unobservability condition, so persistence is not what makes confounding fragile; state-level injection is. Supplement S2 gives two further qualifications (finite-horizon discriminability; identical state effects with different meanings).

For Section 7 this gives a purely linear reason why constructed confounding may not survive in a market, where an incident’s orders propagate through the book. It is a mechanism proved within the linear model, not one isolated in the nonlinear simulator (Section 7.4).

5. Pilot-scale validation in the linear model

We instantiate the model at small scale (a few latent dimensions, fewer observed channels, AR(1) dynamics; parameters in Supplement S2) as a check that the theory is internally consistent and detectable in simulation, not as a market-realistic study. Two regimes are built by the choice of injection directions: *confounded*, with b_K, b_F chosen through the null space of C so that $Cb_K = Cb_F$ exactly, satisfying Theorem 1’s condition; and *separated*, with Cb_K, Cb_F linearly independent. A logistic-regression classifier is trained on whole-episode summary statistics of each observed channel (mean, standard deviation, net drift), in both regimes and at two injection magnitudes.

Table 1: Linear pilot: held-out accuracy and AUC by regime and injection magnitude (480 test episodes per cell).

Regime (Theorem 1’s prediction)	Magnitude	Accuracy	AUC
Confounded (indistinguishable)	0.6	0.529	0.519
Separated (distinguishable)	0.6	0.938	0.983
Confounded (indistinguishable)	1.2	0.487	0.485
Separated (distinguishable)	1.2	1.000	1.000

As Theorem 1 predicts, accuracy in the confounded regime sits at chance regardless of injection magnitude: a more severe incident is not easier to classify when the two observation laws are identical. In the separated regime accuracy is high and improves with magnitude, as Corollary 2 predicts through the whitened separation. Three further classifier families (a random forest, Gaussian naive Bayes and a single-feature threshold rule) give the same pattern (Supplement S2), as Theorem 1 implies for any passive classifier.

6. Cost of ambiguity: definitions, estimator and the linear pilot

Terminology. Three risks are kept separate, in the order $R_{\text{oracle}} \leq R_Y^* \leq R_{\text{const}}$: the true-cause oracle $R_{\text{oracle}} = \mathbb{E}[\min_a L(a, M)]$, the best policy using the observation $R_Y^* = \inf_{\pi} \mathbb{E}[L(\pi(Y), M)]$, and the best constant action

$R_{\text{const}} = \min_a \mathbb{E}[L(a, M)]$. Exact confounding is sufficient, not necessary, for $R_Y^* = R_{\text{const}}$: an informative Y can still leave the optimal action unchanged, for example at a low prior where every posterior stays below the containment threshold. An absence of measured cost benefit is therefore not evidence that two observation laws are equal.

Prior-shift correction of the threshold. The classifier is trained on balanced data, so its score $s(y)$ estimates $P(K \mid Y = y)$ under the training prior $p_{\text{train}} = \frac{1}{2}$, not the deployment prior p . Applying the deployment threshold $q^* = \text{DISRUPT}/(\text{HARM} - C_{OP} + \text{DISRUPT})$ to this raw score is the prior-shift error documented for cost-sensitive learning [20, Sec. 3, Thm. 2]; see also [21]. We apply Bayes’ rule, $q_p(y) = \frac{(p/p_{\text{train}})s(y)}{(p/p_{\text{train}})s(y) + \frac{1-p}{1-p_{\text{train}}}(1-s(y))}$, and threshold q_p at q^* . Without the correction, a classifier under confounding appears worse than ignoring the observation, an artifact of the threshold rather than of confounding. The correction trades false positives for false negatives rather than removing error: at low priors it makes the policy conservative about calling a compromise (Supplement S4). It is more accurate about cost for a given loss table, not uniformly more accurate.

Linear pilot. With an illustrative toy triple $(C_{OP}, \text{DISRUPT}, \text{HARM})$ satisfying Proposition 1’s hypotheses (Supplement S2), we compare four policies across a grid of priors $P(K)$: always contain, always leave, the prior-corrected classifier, and the oracle.

Table 2: Linear pilot, confounded regime: expected cost per episode (toy units). Under exact confounding $R_Y^* = R_{\text{const}}$, so the last column is the classifier’s excess over the best achievable cost: mean over seeds, with the range over seeds in brackets. Per-seed values are test-set estimates and can fall on either side of the floor.

$P(K)$	Contain	Leave	Classifier (sd)	Oracle	R_{const}	Classifier $- R_{\text{const}}$ [min, max]
0.05	5.850	0.500	0.500 (0.000)	0.150	0.500	0.000 [0.000, 0.000]
0.1	5.700	1.000	1.000 (0.000)	0.300	1.000	0.000 [0.000, 0.000]
0.2	5.400	2.000	2.000 (0.000)	0.600	2.000	0.000 [0.000, 0.000]
0.3	5.100	3.000	3.000 (0.000)	0.900	3.000	0.000 [0.000, 0.000]
0.5	4.500	5.000	4.520 (0.085)	1.500	4.500	0.020 [−0.125, 0.196]

The constant and oracle policies match their closed forms $pC_{OP} + (1 - p)\text{DISRUPT}$, $p\text{HARM}$ and pC_{OP} , a direct check of the cost computation. Under confounding no policy of Y has expected cost below $R_Y^* = R_{\text{const}}$, so the fitted classifier can at best reach this floor, and it did so exactly at every prior below $q^* = 0.462$: in all 20 seeds it took the best constant action

on every test episode. At the highest prior, the only one above q^* and the closest to it, the fitted score straddles the threshold and the classifier left some episodes uncontained; its cost then scatters around the floor (mean excess 0.020, seed standard deviation 0.085; 9 of 20 seeds at or below it), and seeds below the floor reflect finite test sets, not a gain. Proposition 1 concerns the Bayes-optimal policy; it implies that a fitted classifier cannot beat the floor, not that it attains it. Under separation the classifier beats the best constant policy (Supplement S2, with a sensitivity analysis over cost triples).

6.1. Where the dollar cost triple comes from

For the ABIDES study we reconstruct a dollar cost triple $(C_K, C_F, H_K) = (C_{OP}, \text{DISRUPT}, \text{HARM})$ from the Almgren–Chriss market-impact model [25] in its continuous-trading form, the expected shortfall of a uniform-rate liquidation, $L(X, T) = \varepsilon X + (\frac{\gamma}{2} + \frac{\eta}{T})X^2$: their expression with the discrete-interval correction $\tilde{\eta} = \eta - \gamma \Delta t/2$, for a trading interval Δt , taken in the limit $\Delta t \rightarrow 0$, where $\tilde{\eta} = \eta$. The dropped term is largest for a single interval spanning the whole window; with the model’s parameters it is then 0.064% of η for the urgent containment window T_u , 0.38% for the harm window and 2.3% for the patient horizon T_p , the longest window of the triple and the one that attains the bound. The triple uses the model’s own illustrative parameters applied to three scenarios: a routine containment trade, a benign order forced into an urgent unwind, and an unmitigated adversarial sequence. This gives $(C_K, C_F, H_K) = (\$490,938, \$473,958, \$1,317,500)$, which satisfies $H_K > C_K$. It is an arithmetic illustration, not a calibration to any venue or incident; the costs are not measured economic losses (worked example in Supplement S4).

6.2. A positive fault loss

The simulated fault mechanisms submit real orders, so leaving a fault alone is not free. We replace the zero entry by H_F , the Almgren–Chriss cost of the fault mechanism’s directly measured submitted volume in each episode, averaged over episodes (so the convexity of L in X is included); it is a modelled cost of submitted, not executed, volume, measured on the same episodes as the classification cells (Section 7.4). Per-design values are in

Supplement S4; they range from \$850 to \$104,739 per episode. With $H_F > 0$,

$$R_{\text{oracle}} = p C_K + (1 - p) H_F,$$

$$R(\pi) - R_{\text{oracle}} = p (H_K - C_K) \text{FNR} + (1 - p) (C_F - H_F) \text{FPR},$$

and the cost-optimal threshold on the prior-corrected score is $q^* = (C_F - H_F) / [(H_K - C_K) + (C_F - H_F)]$, which depends on the loss table only (between 0.309 and 0.364 across designs and regimes). The zero- H_F formulas above hold for the toy scenario only. Each cell’s cost is estimated by direct class-conditional weighting, $\hat{R}_p = p\{C_K + (H_K - C_K)\hat{\text{FNR}}\} + (1 - p)\{H_F + (C_F - H_F)\hat{\text{FPR}}\}$, from the held-out error rates at q^* , with no resampling step.

Two fault-cost conventions. Both conventions share the compromise column, $C_K = L(X_0, T_u)$ and H_K (Section 6.1), with X_0 the reference order of that section and T_u its urgent horizon, and the same $H_F = \frac{1}{N} \sum_i L(X_i, T_n)$: the cost of each fault episode’s measured volume X_i executed at the pace it actually ran (the post-onset window, T_n). They differ only in C_F .

- *Current convention (primary).* $C_F = \text{DISRUPT} = L(X_0, T_u) - L(X_0, T_p)$, with T_p the patient horizon: the premium of forcing the reference benign order into an urgent rather than a patient unwind, one value for every cell. It prices a kill-switch under a fault as disruption of benign business, independent of the fault’s volume; C_F is incremental while H_F is absolute.
- *Rebaselined convention (sensitivity arm).* $C_F = \frac{1}{N} \sum_i L(X_i, T_u)$: containing a fault means unwinding the fault’s own volume urgently. Both fault entries are then absolute costs of the same volume, differing only in urgency, so $C_F - H_F$ is an urgency premium; C_F varies by design and regime (Supplement S4).

Which convention models a gateway kill-switch under a fault is a modelling choice, not a data question. We report the current convention as primary, as fixed in the analysis plan before the table was computed, and the rebaselined convention as a sensitivity arm.

The policy comparison under this loss is reported once, in Section 7.3.

7. Observation matching in an agent-based market

7.1. Scope and constructions

What this section does and does not test. Theorem 1 concerns equality of the whole observation-trajectory law, $P_K(Y_{0:T}) = P_F(Y_{0:T})$. The

constructions below match the two mechanisms on one or a few chosen summary statistics while other observable channels may still differ. Matching one moment is not equality of the joint law, and a classifier exploiting an unmatched channel does not contradict Theorem 1, even in a linear system: $\mathcal{N}((0, 2), I)$ and $\mathcal{N}((0, -2), I)$ share their first-coordinate mean. No linearization bridge (operating point, Jacobian, approximation-error bound) connects the linear model to the simulator. This section is therefore not a test of Theorem 1; it asks which observable channels leak information under which summary-matching choices, and what that does to decision cost.

We inject compromise and fault mechanisms as order attributes (submission rate, order size, price aggressiveness, directional bias, cancel/replace churn) into ABIDES [26], a discrete-event multi-agent market simulator with a background population of noise, value and market-maker agents. Each design has two regimes. In the *summary-matched* regime a calibration search sets the fault’s attributes so that it matches the compromise on a chosen target statistic; in the *contrast* regime the two mechanisms are deliberately given a large difference in a chosen attribute. The classifier (logistic regression) uses whole-episode summaries of the observed channels (Section 3.5).

Five profiles. The *original design* varies all five attributes and calibrates on mean price drift, with four observed channels (spread, depth, cancellation rate, mid-price). *Study 4* reuses the original design’s episodes with three added channels: order-flow imbalance [27], signed executed volume and queue concentration. *Study 5* varies a different attribute subset with a deterministic wake-up clock and a different calibration target. The *third design* calibrates on total order count and leaves the cancellation rate free. The *fourth design* varies directional bias, size and aggressiveness and calibrates on mean spread. The designs differ in which attributes are varied and which statistic is matched, the choices that determine what can leak. We stopped adding designs when the first three had shown three distinct failure modes; a fourth, built to probe that stopping rule, produced a further distinct profile, so the design space is not exhausted (Supplement S3). Study 4 is not a separate simulation: it re-featurizes the original design’s episodes with added channels and shares their fault volume and H_F . Its cells therefore measure the effect of a richer feature set on the same data, not a replication, and 50-cell counts include the original design’s episodes twice. We call the study’s profiles by these five names throughout; the tables’ row labels map to them as stated in the caption of Table 3.

Era-3 basis. Every headline ABIDES result is generated with the third,

hardened version of the simulation pipeline (the *Era-3* pipeline: separate worker processes, one task per worker, and a runtime check that the random-number-generator patches are active; Section 7.4, Supplement S6), on *paired* seeds, each yielding one compromise and one fault episode. Results from the two earlier generation eras are exploratory.

7.2. Classification on the paired basis

Table 3: Classification by design and regime. The ten classification cells are computed on paired seeds: each seed yields one compromise and one fault episode, the train/test split is grouped by seed, and the intervals are cluster-bootstrap intervals over seeds, all under the hardened pipeline (Section 7.4). Study 4 uses the original design’s episodes with three added channels; the design codes V1–V4 in the row labels are used only in the tables. [†] Zero test errors: exact Clopper–Pearson interval over the test episodes (the bootstrap interval is degenerate; Supplement S6).

Design	Contrast		Summary-matched	
	Accuracy [95% CI]	AUC [95% CI]	Accuracy [95% CI]	AUC [95% CI]
Original design (V1)	0.685 [0.652, 0.718]	0.756 [0.728, 0.786]	0.857 [0.830, 0.882]	0.934 [0.917, 0.950]
Study 4 (V1, seven channels)	0.998 [0.995, 1.000]	1.000 [1.000, 1.000]	0.898 [0.875, 0.920]	0.966 [0.954, 0.977]
Study 5 (V2)	0.663 [0.633, 0.692]	0.731 [0.706, 0.757]	0.553 [0.525, 0.580]	0.577 [0.551, 0.604]
Third design (V3)	0.998 [0.995, 1.000]	1.000 [1.000, 1.000]	0.875 [0.850, 0.898]	0.953 [0.940, 0.967]
Fourth design (V4)	0.727 [0.698, 0.757]	0.807 [0.783, 0.832]	1.000 [0.994, 1.000] [†]	1.000 [1.000, 1.000]

Every cell has 1000 episodes per class and 600 held-out test episodes from 300 test seeds. A cell is *at chance* when its accuracy interval contains one half, and *above chance* when the lower end of its accuracy interval is above one half; the interval is the cluster bootstrap over test seeds, or the exact Clopper–Pearson interval for a zero-error cell. An interval containing one half would not establish equal observation laws: a wide interval leaves room for undetected signal.

Table 3 shows no single ranking. In the original design the summary-matched regime is the more classifiable one, with non-overlapping accuracy intervals; with Study 4’s added channels the contrast regime becomes nearly perfectly classifiable, because the contrast mechanisms were always separable on signed executed volume, which the original four channels did not expose (Supplement S3). Study 5 is weakly classifiable in both regimes. In the third design both regimes are strongly classifiable: its mechanisms differ in cancellation rate, one of the observed channels, which matching total order count left free. The fourth design’s summary-matched cell has 0 test errors (exact 95% Clopper–Pearson lower bound 0.994 over test episodes,

0.988 over test seeds): calibrating on mean spread left a large difference in effective order size, which leaks through depth: a size-based channel, not the cancellation-rate channel of the third design. In each of the four injection designs, calibration by search left the summary-matched regime above chance by its 95% interval (Table 3): clearly in the original, third and fourth designs, and marginally in Study 5, whose accuracy interval (0.525–0.580) and AUC interval (0.551–0.604) have lower ends close to one half. The caution stated above for intervals containing one half applies in the other direction too: these are per-cell intervals, not adjusted for the 10 comparisons, and a lower end this close to one half is weak evidence of residual signal. Study 5’s result is therefore marginal; the other three designs’ are not. The four designs are parameterizations of one shared simulation infrastructure (Section 9), so this is a regularity of the constructions tested, not a general law about calibration by search.

The mechanism is structural. In the linear model one vector equality in $\mathbb{R}^m$ removes the difference between mechanisms in every observed coordinate at every time step, because C is fixed and known, and even that equality is fragile under state feedback (Theorem 2). In a multi-agent market there is no fixed linear map to match: a designer matches a finite set of statistics, and whatever is left unmatched can leak. How much leaks depends on the observer’s channels (the rows of C in the linear analog), on the incident’s footprint relative to background noise, and on classifier capacity — all representable, though unexplored, in the linear pilot. What the linear model does not need is calibration by search: its confounded pair has a closed-form construction, whereas the simulator’s map from attributes to statistics has none. None of the summary-matched constructions was matched to a pre-specified tolerance on independent data (Supplement S3), so “summary-matched” means a match attempted, not achieved.

7.3. Cost of ambiguity on the paired basis

The cost table applies the dollar triple of Section 6.1 and the modelled H_F (from measured submitted volume) of Section 6.2 to five profiles $\times$ two regimes $\times$ five priors (50 cells; full table in Supplement S4). Study 4 shares its incident mechanism with the original design and reuses its H_F . The best constant policy’s excess over the oracle agrees with the identity $B(p) = \min\{p(H_K - C_K), (1 - p)(C_F - H_F)\}$ in every cell. Intervals on differences between policies are cluster-bootstrap intervals over test seeds; the constant policies’ costs are not resampled.

At point estimate (strict inequality), the classifier beats always-contain in 50 of 50 cells and the best constant policy in 44 of 50. With cluster-bootstrap intervals over seeds, its advantage over always-contain is a supported improvement in 49 cells and inconclusive at this power in 1; against the best constant it is a supported improvement in 40 cells and inconclusive in 10. No cell shows a supported deterioration against either comparator: the classifier-based policy is never worse than the best constant policy with a supported interval. The 50 cells are not 50 independent comparisons: the five priors of a profile reweight the same test episodes, Study 4 re-featurizes the original design’s episodes, and the two regimes of a design share seeds, so the cells come from 10 profile–regime units drawn from 8 episode sets. Each verdict uses its own 95% interval, not adjusted for the number of cells; the counts describe the table and carry no family-wise error rate. An adjustment for multiplicity only widens intervals, so it cannot create a supported deterioration; it can only reduce the number of supported improvements. By unit (Table 4), the improvement over the best constant is supported at every prior in 6 units, at some priors in 3, and at none in 1, the Study 5 summary-matched unit. The cells without a supported improvement are listed in Table 5. These cells lie at the low end of the prior grid or in the Study 5 profile; in all of them the best constant policy is always-leave, except the Study 5 summary-matched cell at the highest prior, where it is always-contain. Point-estimate counts use strict inequality; Table 5 gives the interval verdicts.

Table 4: Cost verdicts against the best constant policy by profile–regime unit, under the current and the rebaselined fault-cost conventions: number of priors (of five) with a supported improvement, with a point-estimate exception, and the best constant policy by prior. Per-cell 95% cluster-bootstrap intervals, not adjusted for multiplicity; all cells in Supplement S4.

[L8: table_cost_units.tex]

The exceptions are concentrated in Study 5 (5 of the 6 point-estimate exceptions); the other profiles have a supported advantage over the best constant in every cell except a few contrast cells at the low end of the prior grid (Table 5). At the lowest priors of the Study 5 summary-matched profile the classifier’s cost equals the best constant’s exactly: a tie, counted as an exception under strict inequality. On the same contrast episodes, the Study 4 feature set gives a supported improvement over the best constant in every cell, whereas the original feature set leaves the lowest-prior cells inconclusive:

Table 5: Cells whose advantage over the best constant policy is not a supported improvement (paired basis, current fault-cost convention). Costs in dollars (illustrative reconstruction, Section 6.1); difference = classifier minus best constant, with its cluster-bootstrap 95% interval over seeds.

Design	Regime	Prior	R_{clf}	Best constant	Difference [95% CI]	vs best constant	vs always-contain
Original design (V1)	contrast	0.05	99,906	97,510 (leave)	2,395 [-1,791, 7,564]	inconclusive	improvement
Original design (V1)	contrast	0.10	160,009	161,720 (leave)	-1,711 [-7,827, 5,451]	inconclusive	improvement
Study 5 (V2)	contrast	0.05	67,767	66,682 (leave)	1,085 [-827, 4,357]	inconclusive	improvement
Study 5 (V2)	contrast	0.10	128,699	132,515 (leave)	-3,816 [-6,894, 401]	inconclusive	improvement
Study 5 (V2)	summary-matched	0.05	66,682	66,682 (leave)	0 [0, 0]	inconclusive	improvement
Study 5 (V2)	summary-matched	0.10	132,515	132,515 (leave)	0 [0, 0]	inconclusive	improvement
Study 5 (V2)	summary-matched	0.20	266,863	264,180 (leave)	2,683 [-551, 7,178]	inconclusive	improvement
Study 5 (V2)	summary-matched	0.30	400,582	395,845 (leave)	4,737 [-6,297, 15,234]	inconclusive	improvement
Study 5 (V2)	summary-matched	0.50	481,261	482,448 (contain)	-1,187 [-6,308, 4,133]	inconclusive	inconclusive
Fourth design (V4)	contrast	0.05	69,353	71,486 (leave)	-2,133 [-5,648, 2,486]	inconclusive	improvement

a feature-set effect on shared data, not evidence from a second design. Under the rebaselined fault-cost convention (Section 6.2) always-contain is the cheaper constant action in most cells (36 of 50), so the best constant policy changes. Against it, the classifier’s improvement is supported in 34 cells and inconclusive in 16; no cell shows a supported deterioration against either comparator. Its point estimate does not beat the best constant in 11 cells: in the 10 Study 5 cells the classifier contains every test episode and its cost equals always-contain’s exactly, and every other exception is inconclusive. Unlike under the current convention, the cells without a supported improvement are not confined to the low end of the prior grid and Study 5: some lie at the highest prior in other designs (Table 4; all cells in Supplement S4). The location statement above therefore holds under the current convention only; the absence of a supported deterioration against the best constant holds under both. The classifier beats always-contain at point estimate in 39 of these cells.

Under the current convention, the comparison with always-contain is weak evidence on its own: at every prior of the grid below the highest, always-leave is the cheaper constant action, so any policy that mostly leaves alone beats always-contain; in the two lowest-prior Study 5 summary-matched cells the classifier leaves every test episode alone and its supported improvement over always-contain is exactly that of always-leave. The informative comparison is with the best constant policy.

7.4. Checks that bound the claims

Negative control. Two incident variants with identical observable order-generation kernels, differing only in a behaviourally inert label, were run through the full pipeline, with compromise and fault seeds interleaved from

one pool and independent replicate datasets rather than resplits of one dataset. With identical kernels, neither metric shows a signal: over 20 independent replicates both intervals contain one half (accuracy 0.505 [0.466, 0.544], AUC 0.483 [0.444, 0.522]). Equivalence to chance within the pre-specified margin [0.45, 0.55] is shown for accuracy only: the AUC interval extends below the margin because its mean lies below one half, by less than one standard error, so the AUC result is inconclusive at this power rather than evidence of leakage, which would appear as an interval excluding one half. The control checks the pipeline and evaluation boundary, not Theorem 1; no formal two-one-sided-tests procedure was run (Supplement S5).

Provenance of each result. The ten headline classification cells (Table 3) and the cost table built on them were generated with the hardened pipeline (spawn, one task per worker, runtime patch fingerprint passing for every task), on paired seeds. The modelled fault-side harm H_F used by the cost table (from measured submitted volume) is computed per fault episode from those same episodes, whose trajectories were recovered and checked bit-identical to the stored features, not from an earlier run. Every other ABIDES result carries its generation era (Supplement S6); results from the two earlier eras were produced before the random-number-generator (RNG) patches or with scripts that would not detect a mid-run patch, are reported as exploratory, and not all of their seed blocks are in the provenance register.

Persistence ablation (exploratory). In one run per condition, made before the hardened pipeline, removing persistence lowered classifiability in both scenarios, by a larger margin in the summary-matched scenario than in the contrast scenario, consistent with the duration and dose confound of that design; we do not attribute the difference to any isolated mechanism, and it is not a test of Theorem 2. Supplement S5 describes it, with the isolation redesign (inconclusive), leave-one-design-out on the paired basis (the separating direction does not transfer across designs; a robustness check, not a claim), the signed-volume convention and the error-rate decomposition; Supplement S3 reports the earlier-era secondary studies (seed and classifier-family robustness, population scale, per-feature isolation, footprint sweep).

7.5. Outlook: an action on a second information channel

In the cells of Table 5 (Section 7.3), the fitted classifier’s advantage over the best constant policy is not a supported improvement: passive observation adds no supported value there. Whether an action on a different information channel can add value in those cells is outside this paper. A companion paper

in preparation [28] asks, for the cells of this set in which the classifier’s point estimate does not beat the best constant policy, whether targeted verification does, and over which region of the verification parameters. It examines this as an exploratory test. This paper claims no outcome for that test, and does not claim that these cells are where verification’s effect is largest.

Two results bound what this paper contributes to that question. Whether K and F are confounded (Theorem 1) is a property of the mechanism instances and the observation law, not of the prior p ; what depends on p is the value of resolving the ambiguity, Proposition 1’s $\min\{(1-p)\text{DISRUPT}, p(\text{HARM} - C_{OP})\}$, zero exactly when $p \in \{0, 1\}$. And exact confounding is not generic under state-level injection (Theorem 2), while no ABIDES construction was shown to instantiate Theorem 1’s premise. This paper therefore defines no confounded region for a real incident and evaluates no verification channel.

8. Discussion

The linear identifiability boundary is exact within its scope and remains a legitimate diagnostic wherever that scope holds, plausibly small perturbations around a steady-state book. Its scope is narrower than Theorem 1 alone suggests. Confounding that survives an output-level check need not survive feedback into the dynamics, and the gap is governed by a classical object, the unobservable subspace of (A, C) (Section 4.1). The ABIDES study adds a second, independent caution: engineering a nonlinear analog of exact confounding by matching summary statistics left every summary-matched regime classifiable above chance, through channels that depended on the design. That calibration needed a search is a fact about the four designs tested, not a dividing line between linear and nonlinear systems. A third caution is methodological: a decision-cost result is only as reliable as the decision rule behind it, and a threshold set without prior-shift correction can manufacture a cost finding in either direction (Section 6). For practitioners, Supplement S7 turns these observations into a checklist, restricted to what the paired-basis results and the audits support.

9. Limitations

1. **Retrospective, whole-episode diagnosis.** Every classifier uses complete-episode features; no result measures the value of acting at a real decision deadline (Section 3.5).

2. **One simulator, one shared infrastructure.** The four ABIDES designs are parameterizations of one shared code base (configuration builder, kernel, features, cost pipeline), not independently coded implementations, so a bug common to that infrastructure would not be caught by comparing designs. The five profiles are regularities of the constructions tested; the design space is open-ended. Verdicts are per-cell intervals without multiplicity adjustment; the unit of independent evidence is the design, of which there are four.
3. **Illustrative costs.** The dollar triple is an Almgren–Chriss illustration, not a calibration, and H_F prices submitted, not executed, volume with a market-impact model; neither is a measured economic loss.
4. **Reference policies.** The constant policies are decision-theoretic reference points, not models of deployed kill switches; an operational comparison would need a specified threshold detector with comparable feature access and timing.
5. **Power and degenerate cells.** Of the cells listed in Table 5, the two lowest-prior Study 5 summary-matched cells are exact ties: the fitted policy leaves every test episode alone there, so it reduces to the best constant action. The other listed cells are inconclusive at the sample size used, not shown to be ties. In addition, the persistence ablation is exploratory, one run per condition before the hardened pipeline (Section 7.4).

10. Conclusion

For a linear-Gaussian model of gateway-observable order flow, a compromise and a fault are observationally indistinguishable if and only if their projected injections coincide, and under that confounding no policy using the observation beats the best constant action; the bound extends to approximate confounding in total variation.

Within the same linear model, this confounding is fragile: once the incident propagates through the dynamics, it survives only when the injection difference lies in the Kalman unobservable subspace, which is trivial for generic dynamics.

In the ABIDES simulator, calibration by search did not bring the summary-matched regime to chance in any of the four designs, clearly in three and marginally in Study 5, and which channel leaked depended on the design; the study is about observation matching, not a test of the theorem.

With a positive fault loss modelled from measured submitted volume (Section 6.2), the cost comparison on the paired Era-3 basis is stated in Section 7.3: the classifier-based policy is never worse than the best constant policy with a supported interval, under both fault-cost conventions. Under the current convention the cells without a supported improvement over it lie at the low end of the prior grid or in the Study 5 profile, in one Study 5 cell the best constant is always-contain, and its point-estimate cost is below always-contain’s in every cell; under the rebaselined convention neither the location statement nor the always-contain comparison holds in every cell. The verdicts are per-cell intervals over 10 profile–regime units, not adjusted for multiplicity. The cells in which the fitted classifier adds no supported value over the best constant policy (Section 7.3) leave a question to a companion paper in preparation, stated in Section 7.5 together with what this paper does not claim about it. Whether any of this value is available at an operational decision deadline is a separate question, which needs prefix features and is not answered here.

Data and code availability

The simulation code, the experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). Until then they are available from the corresponding author on reasonable request. The ABIDES market simulator is open-source software; the patches applied to it are part of the deposit.

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used large-language-model assistants (Claude, Anthropic) to draft and revise parts of the text and to write and review parts of the analysis code. After using these tools, the author reviewed, checked and edited the content as needed and takes full responsibility for the content of the published article.

Appendix A. Proofs of the main results

This appendix gives the proofs of the three results on which the title and Contributions 1–2 rest. The proofs of Corollaries 1–3, Propositions 2–3 and Theorem 3 are in Supplement S1, which also repeats these three.

Appendix A.1. Proof of Theorem 1

Proof. Work in the output-level injection model (Definition D2) with window $W = \{t : t \geq 0\}$ (Definition D3); by (A4) the onset τ is deterministic and common to both instances (Definition D1), so (A5) holds for this W .

Step 1 (decomposition). Under Definition D2 the state recursion does not see the injection, so $x_t = A^t x_0 + \sum_{s=0}^{t-1} A^{t-1-s} w_s$ under either instance. Hence, for every $t \geq 0$,

$$y_t = G_t + m_M(t), \quad G_t = Cx_t + \varepsilon_t, \quad m_M(t) = \mathbb{K}[t \geq \tau] \theta_M Cb_M.$$

The process $G = (G_t)_{t \geq 0}$ is a fixed measurable function of $(x_0, \{w_t\}, \{\varepsilon_t\})$ through A and C . By (A1) and (A2) the law of $(x_0, \{w_t\}, \{\varepsilon_t\})$ does not depend on the instance, and by (A3) neither do A and C ; therefore the law of G is the same under both instances. The sequence m_M is deterministic (Definition D1, (A4)).

Step 2 (if). Suppose $\theta_K Cb_K = \theta_F Cb_F$. Then $m_K = m_F =: m$, and under either instance $Y_W = G + m$. For every finite $S \subseteq W$, the law of $(y_t)_{t \in S}$ is the image of the (common) law of $(G_t)_{t \in S}$ under the translation by $(m(t))_{t \in S}$, hence the same under both instances. Laws on the product σ -algebra that

agree on all finite-dimensional marginals are equal (Definition D3), so $P_K^W = P_F^W$.

Step 3 (only if). Suppose $P_K^W = P_F^W$. Each G_t is Gaussian by (A1) and (A2), so y_t is integrable and equal laws give equal means: $\mathbb{E}_K[y_t] = \mathbb{E}_F[y_t]$ for all $t \in W$. Since $\mathbb{E}[G_t]$ does not depend on the instance (Step 1), $m_K(t) = m_F(t)$ for all $t \in W$. Taking any $t \in W$ with $t \geq \tau$ (one exists by (A5); for a non-integer or negative τ take any integer $t \geq \max(\tau, 0)$) gives $\theta_K Cb_K = \theta_F Cb_F$, i.e. (*).

Remarks. (i) The covariance of Y_W is that of G , which does not depend on the instance; this is the only role of the noise. (ii) Gaussianity is used only for integrability in Step 3; Step 2 uses only that the law of G is instance-independent. (iii) If W contained no $t \geq \tau$, that is, if (A5) failed, both laws would always coincide and the “only if” direction would fail; this is why (A5) is listed. $\square$

Appendix A.2. Proof of Proposition 1

Proof. Use this paper’s loss table (Definition D6): $L(\text{contain}, K) = C_{OP}$, $L(\text{contain}, F) = \text{DISRUPT}$, $L(\text{leave alone}, K) = \text{HARM}$, $L(\text{leave alone}, F) = 0$, and the decision problem of Definition D7 on the window W of Theorem 1.

Best achievable cost. By (*) and Theorem 1, $P_K^W = P_F^W =: P$. For a policy π (randomized allowed, Definition D7) let $r = \int \pi dP \in [0, 1]$. Since both expectations in $R(\pi)$ are taken under the same law P ,

$$R(\pi) = r [p C_{OP} + (1 - p) \text{DISRUPT}] + (1 - r) p \text{HARM},$$

an affine function of r . Its infimum over $r \in [0, 1]$ is attained at an endpoint, and both endpoints are attained by constant deterministic policies ($\pi \equiv 1$: always contain; $\pi \equiv 0$: always leave alone). Hence

$$R^* = \min\{p C_{OP} + (1 - p) \text{DISRUPT}, p \text{HARM}\},$$

and the value is the same whether the policy class is deterministic or randomized (the class of Definition D7 used in the statement).

Oracle. By Definition D7, $R_{\text{oracle}} = p \min\{C_{OP}, \text{HARM}\} + (1 - p) \min\{\text{DISRUPT}, 0\} = p C_{OP}$, using $\text{HARM} > C_{OP}$ and $\text{DISRUPT} > 0$.

Cost of ambiguity. Subtracting $p C_{OP}$ inside the minimum,

$$\text{CoA}^* = \min\{(1 - p) \text{DISRUPT}, p(\text{HARM} - C_{OP})\}.$$

Both terms are strictly positive because $0 < p < 1$, $\text{DISRUPT} > 0$ and $\text{HARM} > C_{OP}$; hence $\text{CoA}^* > 0$. $\square$

Appendix A.3. Proof of Theorem 2

Proof. Work in the state-level injection model (Definition D9) with window $W = \{t : t \geq \tau\}$ (Definition D3); by (A4) the onset τ is deterministic and common to both instances (Definition D1). Write $S_k = \sum_{j=0}^k A^j$ for $k \geq 0$.

Step 1 (noise part and deterministic part). Define the uninjected process $z_0 = x_0$, $z_{t+1} = Az_t + w_t$ for all $t \geq 0$, and the deterministic sequence $s_t = 0$ for $t \leq \tau$, $s_{t+1} = As_t + A\theta_M b_M$ for $t \geq \tau$. Then $x_t = z_t + s_t$ for all t : this holds for $t \leq \tau$ because the recursion before the onset is uninjected, and if it holds at some $t \geq \tau$, then by Definition D9

$$x_{t+1} = A(x_t + \theta_M b_M) + w_t = (Az_t + w_t) + (As_t + A\theta_M b_M) = z_{t+1} + s_{t+1}.$$

Unrolling, $s_{\tau+k} = \sum_{j=1}^k A^j \theta_M b_M$, so $\tilde{x}_{\tau+k} = z_{\tau+k} + S_k \theta_M b_M$ and, for every $k \geq 0$,

$$y_{\tau+k} = G_{\tau+k} + m_M(\tau+k), \quad G_t = Cz_t + \varepsilon_t, \quad m_M(\tau+k) = CS_k \theta_M b_M.$$

Step 2 (why the noise does not matter). $G = (G_t)_{t \geq 0}$ is a fixed measurable function of $(x_0, \{w_t\}, \{\varepsilon_t\})$ through A and C and does not involve τ or the instance. By (A1), (A2) and (A3) its law, in particular its covariance, is the same under both instances. The instance enters only through the deterministic mean shift m_M .

Step 3 (laws equal iff shifts equal). If $m_K(t) = m_F(t)$ for all $t \in W$, then $Y_W = G_W + m$ under both instances, so every finite-dimensional marginal of Y_W is the same translate of the common marginal of G , and $P_K^W = P_F^W$ (Definition D3). Conversely, if $P_K^W = P_F^W$, then since each G_t is Gaussian by (A1) and (A2), and hence integrable, $\mathbb{E}_K[y_t] = \mathbb{E}_F[y_t]$ and so $m_K(t) = m_F(t)$ for every $t \in W$. With $\Delta_0 = \theta_K b_K - \theta_F b_F$, $m_K(\tau+k) - m_F(\tau+k) = CS_k \Delta_0$. Hence

$$P_K^W = P_F^W \iff CS_k \Delta_0 = 0 \text{ for all } k \geq 0.$$

Step 4 (telescoping). For every $N \geq 0$: $CS_k \Delta_0 = 0$ for $k = 0, \dots, N \iff CA^k \Delta_0 = 0$ for $k = 0, \dots, N$. Indeed $CS_0 = C$ and $CS_k - CS_{k-1} = CA^k$ for $k \geq 1$, so each family is obtained from the other by taking successive differences or partial sums. Hence $P_K^W = P_F^W \iff CA^k \Delta_0 = 0$ for all $k \geq 0$.

Step 5 (Cayley–Hamilton). Let $\chi(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_0$ be the characteristic polynomial of A . By the Cayley–Hamilton theorem $A^n = -\sum_{i=0}^{n-1} c_i A^i$; multiplying by A^{k-n} and inducting on k , every A^k with $k \geq$

n is a linear combination $\sum_{i=0}^{n-1} \alpha_{k,i} A^i$. Therefore, if $CA^i \Delta_0 = 0$ for $i = 0, \dots, n-1$, then $CA^k \Delta_0 = \sum_i \alpha_{k,i} CA^i \Delta_0 = 0$ for every $k \geq n$; the converse is immediate. So

$$P_K^W = P_F^W \iff \Delta_0 \in \bigcap_{k=0}^{n-1} \ker(CA^k) = \mathcal{N}(A, C)$$

(Definition D10), which is the claim.

Remark. Adding pre-onset times to W changes nothing, since $m_K(t) = m_F(t) = 0$ for $t < \tau$. $\square$

References

- [1] E. Miehling, M. Rasouli, D. Teneketzis, A POMDP approach to the dynamic defense of large-scale cyber networks, *IEEE Transactions on Information Forensics and Security* 13 (10) (2018) 2490–2505. doi:10.1109/TIFS.2018.2819967.
- [2] S. Zonouz, H. Khurana, W. H. Sanders, T. M. Yardley, RRE: A game-theoretic intrusion response and recovery engine, *IEEE Transactions on Parallel and Distributed Systems* 25 (2) (2014) 395–406. doi:10.1109/TPDS.2013.211.
- [3] K. Hammar, R. Stadler, Intrusion prevention through optimal stopping, *IEEE Transactions on Network and Service Management* 19 (3) (2022) 2333–2348. doi:10.1109/TNSM.2022.3176781.
- [4] K. Hammar, R. Stadler, Learning near-optimal intrusion responses against dynamic attackers, *IEEE Transactions on Network and Service Management* 21 (1) (2024) 1158–1177. doi:10.1109/TNSM.2023.3293413.
- [5] K. Hammar, T. Li, R. Stadler, Q. Zhu, Adaptive security response strategies through conjectural online learning, *IEEE Transactions on Information Forensics and Security* 20 (2025) 4055–4070. doi:10.1109/TIFS.2025.3558600.
- [6] L. B. Booker, S. A. Musman, A model-based, decision-theoretic perspective on automated cyber response, *AAAI-20 Workshop on Artificial Intelligence for Cyber Security (AICS)* (2020). arXiv:2002.08957.

- [7] F. Pasqualetti, F. Dörfler, F. Bullo, Attack detection and identification in cyber-physical systems, *IEEE Transactions on Automatic Control* 58 (11) (2013) 2715–2729. doi:10.1109/TAC.2013.2266831.
- [8] T. Roy, S. Dey, On distinguishability of anomalies as physical faults or actuation cyberattacks, *ASME Letters in Dynamic Systems and Control* 4 (3) (2024) 031003. doi:10.1115/1.4065292.
- [9] M. Taheri, K. Khorasani, I. Shames, N. Meskin, Cyberattack and machine-induced fault detection and isolation methodologies for cyber-physical systems, *IEEE Transactions on Control Systems Technology* 32 (2) (2024) 502–517. doi:10.1109/TCST.2023.3324870.
- [10] S. Zhou, C. Liu, J. Lan, X. Ren, X. Wang, Optimal distinguishable detection of faults and attacks in interconnected systems: A Stackelberg game approach, *Automatica* 193 (2026) 113226.
- [11] A. Anwar, A. N. Mahmood, Z. Shah, A data-driven approach to distinguish cyber-attacks from physical faults in a smart grid, in: *Proceedings of the 24th ACM International Conference on Information and Knowledge Management (CIKM)*, 2015, pp. 1811–1814. doi:10.1145/2806416.2806648.
- [12] N. Müller, K. Bao, J. Matthes, K. Heussen, CyPhERS: A cyber-physical event reasoning system providing real-time situational awareness for attack and fault response, *Computers in Industry* 151 (2023) 103982. doi:10.1016/j.compind.2023.103982.
- [13] F. Ma, Y. Dang, Y. Liu, Identifiability-aware, cost-aware triage of physical faults and measurement-integrity anomalies in energy cyber-physical systems, *Sensors* 26 (17) (2026) 5585. doi:10.3390/s26175585.
- [14] A. A. Albustami, A. F. Taha, E. Bou-Harb, Unmasking stealthy attacks on nonlinear DAE models of power grids, *International Journal of Electrical Power & Energy Systems* 167 (2025) 110569. doi:10.1016/j.ijepes.2025.110569.
- [15] R. Alisic, S. Amin, How sensor attacks transfer across Lie groups, preprint (2026). arXiv:2604.03461.

- [16] P. Chullamonthon, P. Tangamchit, Ensemble of supervised and unsupervised deep neural networks for stock price manipulation detection, *Expert Systems with Applications* 220 (2023) 119698. doi:10.1016/j.eswa.2023.119698.
- [17] C. Yagemann, S. P. Chung, E. Uzun, S. Ragam, B. Saltaformaggio, W. Lee, On the feasibility of automating stock market manipulation, in: *Proceedings of the 36th Annual Computer Security Applications Conference (ACSAC)*, 2020, pp. 277–290. doi:10.1145/3427228.3427241.
- [18] S. J. Grossman, J. E. Stiglitz, On the impossibility of informationally efficient markets, *The American Economic Review* 70 (3) (1980) 393–408.
- [19] R. Xu, T. Zariphopoulou, L. Zhang, Decision making under costly sequential information acquisition: The paradigm of reversible and irreversible decisions, *arXiv preprint* (2024). arXiv:2401.00569v4.
- [20] C. Elkan, The foundations of cost-sensitive learning, in: *Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI)*, 2001, pp. 973–978.
- [21] M. Saerens, P. Latinne, C. Decaestecker, Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure, *Neural Computation* 14 (1) (2002) 21–41. doi:10.1162/089976602753284446.
- [22] D. M. Green, J. A. Swets, *Signal Detection Theory and Psychophysics*, Wiley, New York, 1966.
- [23] H. V. Poor, *An Introduction to Signal Detection and Estimation*, 2nd Edition, Springer, 1994.
- [24] L. Le Cam, *Asymptotic Methods in Statistical Decision Theory*, Springer, New York, 1986.
- [25] R. Almgren, N. Chriss, Optimal execution of portfolio transactions, *Journal of Risk* 3 (2) (2001) 5–39. doi:10.21314/JOR.2001.041.
- [26] D. Byrd, M. Hybinette, T. H. Balch, ABIDES: Towards high-fidelity multi-agent market simulation, in: *Proceedings of the 2020 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS)*, 2020, pp. 11–22. doi:10.1145/3384441.3395986.

- [27] R. Cont, A. Kukanov, S. Stoikov, The price impact of order book events, *Journal of Financial Econometrics* 12 (1) (2014) 47–88. doi:10.1093/jjfinec/nbt003.
- [28] C. Mejrissi, Verification as a costed action: When checking beats containing under attack–fault ambiguity, working paper (2026).